

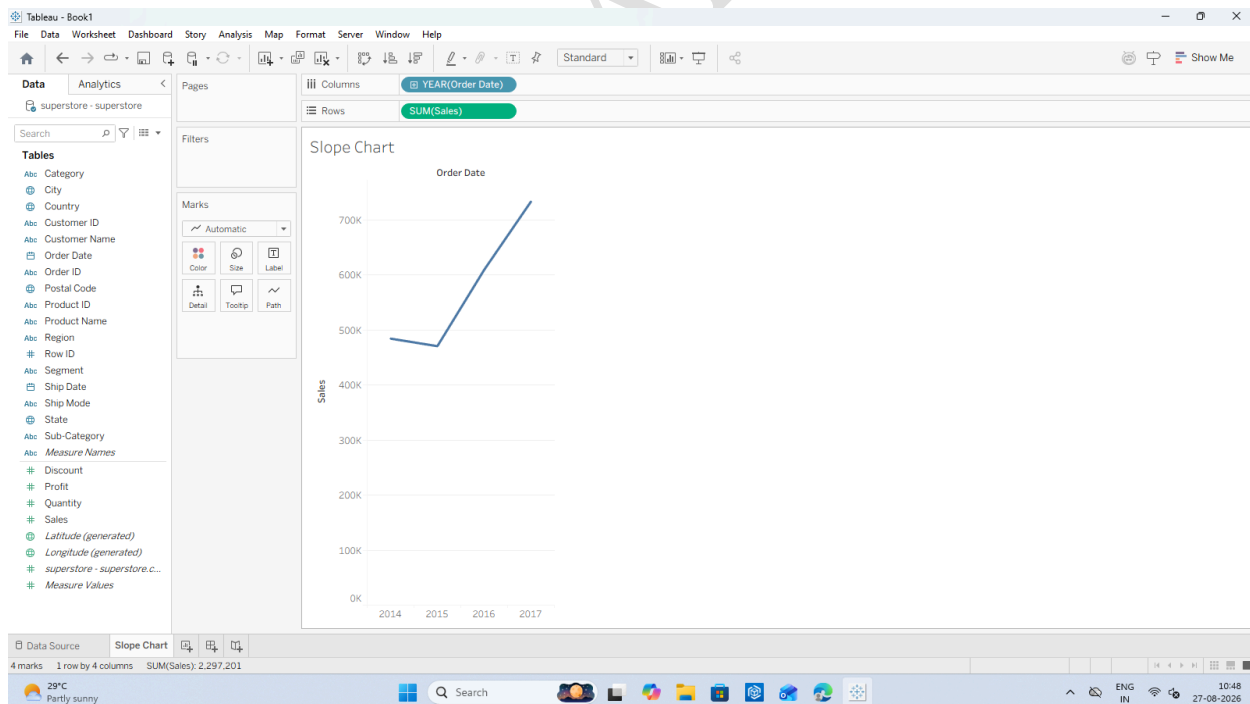
Practical No. 11

Aim: Implementing Advanced Visualization Techniques and Customizations.

- Create slope charts and lollipop charts
- Design waterfall charts
- Build sparklines for trend analysis
- Implement sheet swapping for dynamic dashboards
- Create custom geographic territories
- Use background images in dashboards
- Apply animation and transparency techniques

Dataset: Superstore.csv

- Step 1: Set Up the Base Worksheet 1. Open Tableau Desktop connected to Superstore.csv.
- Create a new worksheet and rename it: Slope Chart.
 - Drag Order Date to the Columns shelf. (It will appear as YEAR(Order Date)).
 - Drag Sales to the Rows shelf (appears as SUM(Sales)).



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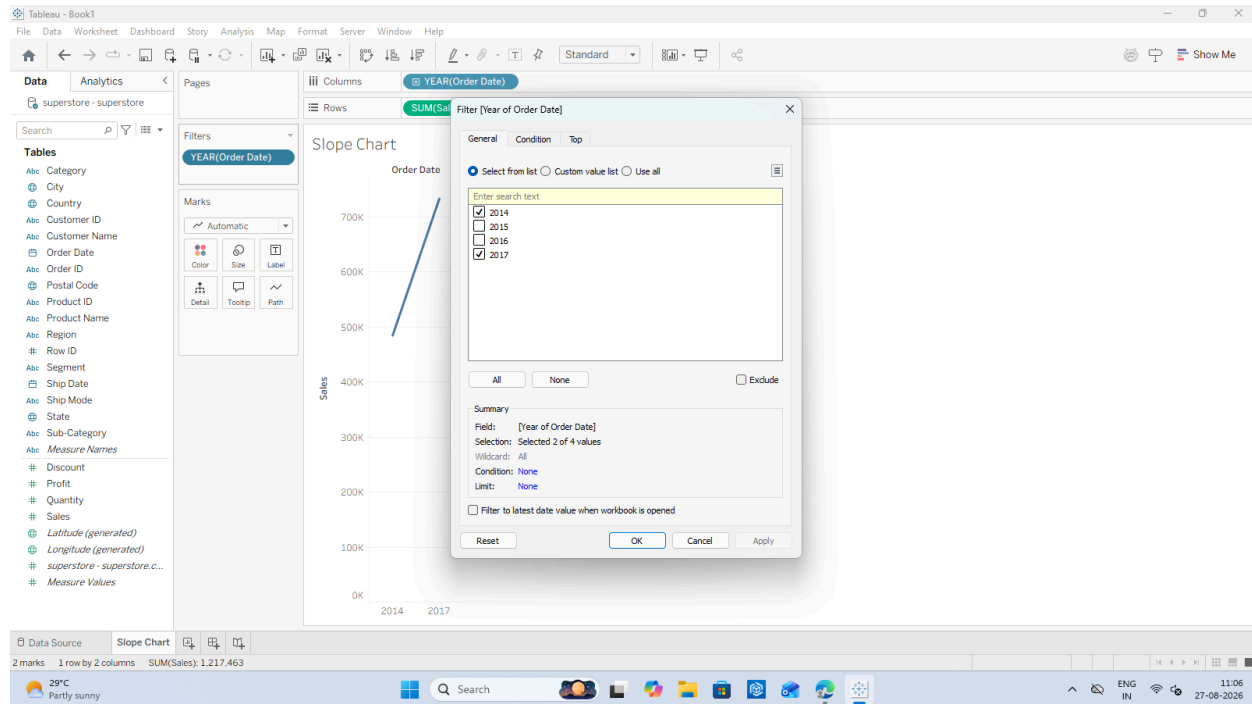
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Step 2: Filter to Baseline & Comparison Years

1. Hold Ctrl , drag Order Date to the Filters shelf, select Years, and click Next.
2. Check only the first year (2014) and the last year (2017). Click OK.
3. Right-click on the YEAR(Order Date) pill on the Columns shelf --> select Discrete (blue pill) so it acts as two distinct category pillars.



Step 3: Disaggregate by Sub-Category

1. On the Marks card, change the mark type dropdown from Automatic to Line.
2. Drag Sub-Category onto the Detail tile (and onto the Color tile) on the Marks card.
3. You will now see 17 distinct sloping lines connecting 2014 to 2017.

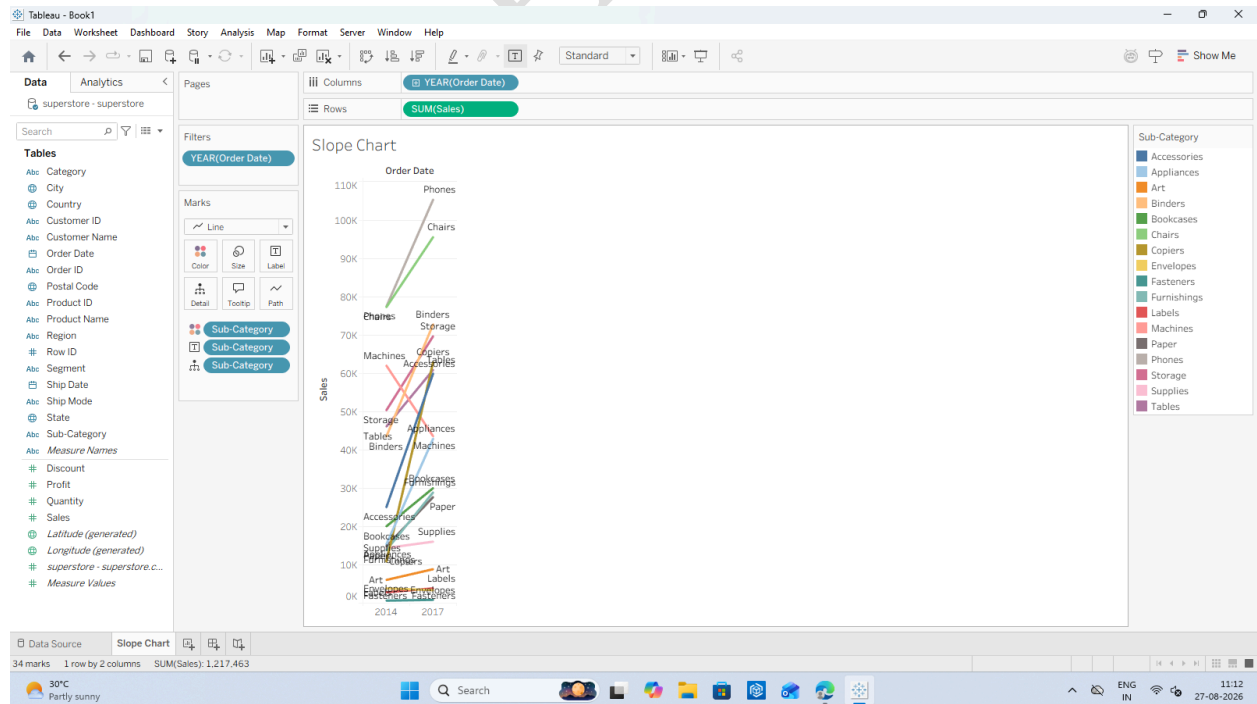
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The screenshot displays the Tableau Desktop interface with a Slope Chart visualization. The chart is titled "Slope Chart" and "Order Date". The Y-axis is labeled "Sales" and ranges from 0K to 110K. The X-axis shows the years 2014 and 2017. The chart displays multiple lines representing different sub-categories, showing their sales trends over time. The legend on the right lists the sub-categories: Accessories, Appliances, Art, Binders, Bookcases, Chairs, Copiers, Envelopes, Fasteners, Furnishings, Labels, Machines, Paper, Phones, Storage, Supplies, and Tables. The chart shows that most sub-categories show an upward trend in sales over the four-year period, with "Accessories" and "Appliances" showing the highest sales values.

1. On the Marks card, click the Label tile:

2. Drag Sub-Category onto Label on the Marks card to display the product name directly at the line endpoints.



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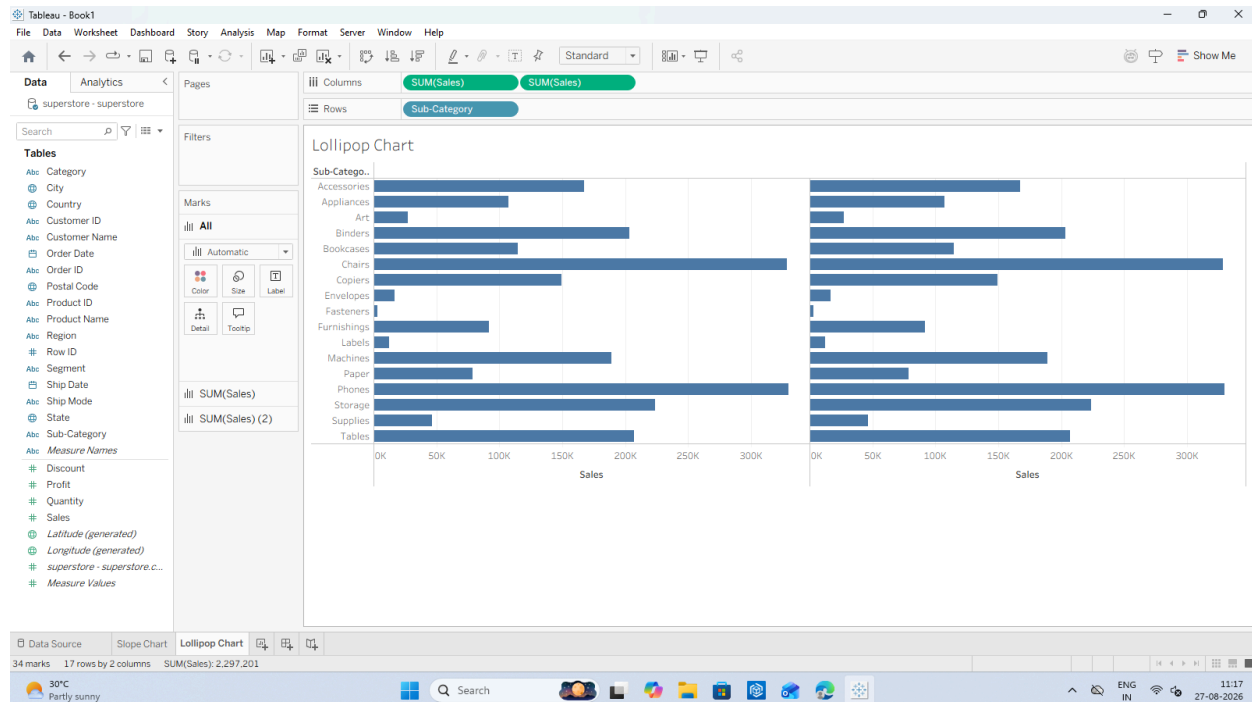
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2. Create a Lollipop Chart (Sales with Profit Highlighting)

Step 1: Set Up the Base Worksheet

1. Create a new worksheet and rename it: Lollipop Chart.
2. Drag Sub-Category onto the Rows shelf.
3. Drag Sales onto the Columns shelf.
4. Drag Sales onto the Columns shelf a second time (creating two adjacent SUM(Sales) pills on Columns).



Step 2: Configure the Marks Cards In the Marks pane on the left, you will now see three cards: All, SUM(Sales) (first mark), and SUM(Sales) (2) (second mark).

1. Configure the Stem (Bar):
 - o Click on the first SUM(Sales) mark card.
 - o Change the mark type dropdown to Bar.
 - o Click the Size tile --> pull the slider down until the bars become thin, elegant Stems.
 - o Click Color --> set the bar color to a neutral gray (e.g., #B0B0B0).
2. Configure the Head (Circle):
 - o Click on the second SUM(Sales) (2) mark card.
 - o Change the mark type dropdown to Circle.
 - o Click the Size tile --> increase the circle size so it sits neatly at the tip of each bar stem.
 - o Drag Profit onto the Color tile of this second marks card.
 - o (Observation): Sub-categories like Tables immediately show negative profit (red/orange) while having high sales.

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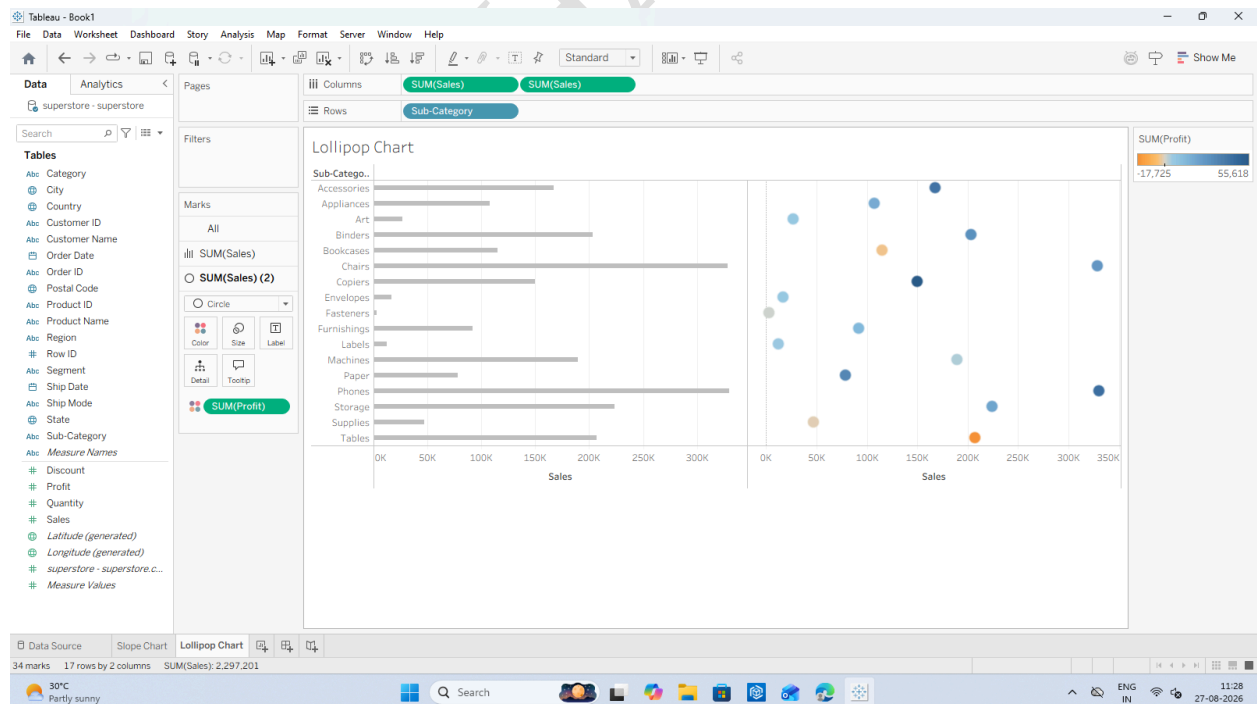
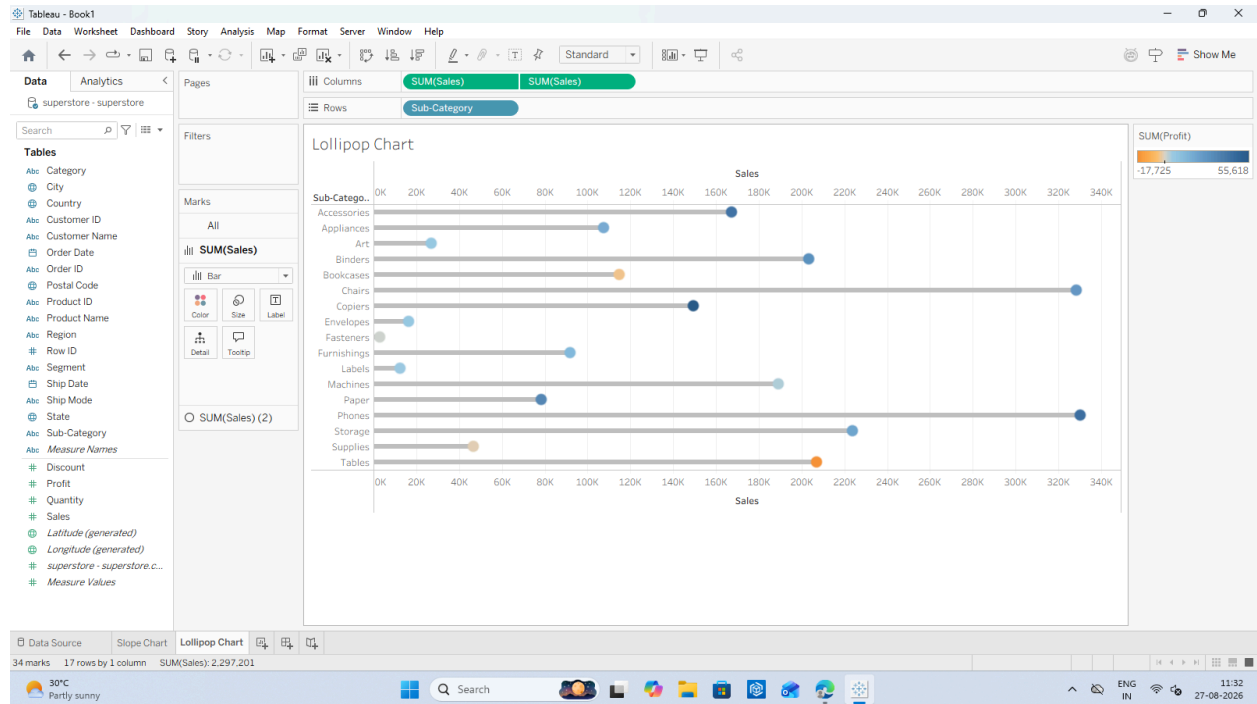
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Step 2: Convert to Dual Axis

1. Right-click the second SUM(Sales) pill on the Columns shelf --> select Dual Axis.
2. Right-click the top numerical axis in the visualization view --> select Synchronize Axis (crucial to ensure both axes align).



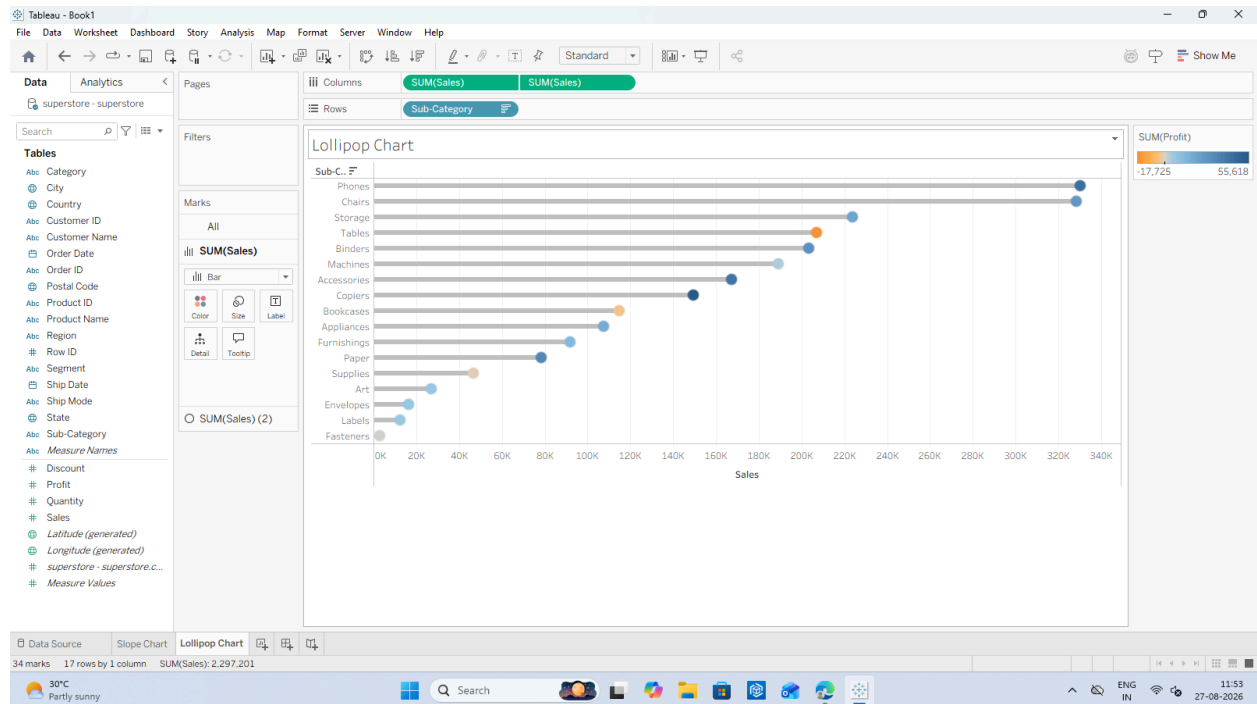
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Step 4: Clean Up Axis Formatting

1. Right-click the top header axis --> uncheck Show Header.
2. Click the Sort Descending button in the top toolbar to sort by total sales.



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